

Applied Section

Combining generalized linear models and credibility models in practice

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Rating of non-life insurance contracts commonly employs multiplicative models, which are estimated by generalized linear models (GLMs); another useful tool for rate making are credibility models. The object of this paper is to demonstrate how these can be combined in practice, to solve the problem with multi-level factors – rating factors with too many levels for GLM estimation. In particular, we consider *car model classification* in motor insurance, using data from a Swedish insurance company.

Keywords: Generalized linear models; Credibility theory; Hierarchical credibility; Car model classification; Motor insurance; Multi-level factor

1. Introduction

In order to rate non-life insurance contracts, actuaries commonly use multiplicative models, where a number of rating factors are estimated by generalized linear models (GLMs). Another useful tool for pricing is credibility theory, however, in our experience credibility is not used in practice as much as it deserves, especially in the private lines. Traditional texts on credibility theory give little or no guidance for the practical use of credibility estimators in a multiplicative GLM environment. The object of this article is to show how GLMs and credibility can be combined in practice, with special focus on the problem of *car model classification* for motor insurance, and to present an application on data from a Swedish insurance company. We call this combination GLMs with credibility (GLMC).

Dannenburg *et al.* (1996) states that a fact that “hampers the acceptance of credibility techniques by practitioners” is that “credibility is currently taught in a needlessly complicated way” and that credibility is “set in a Bayesian framework”. They suggest replacing the Bayesian setting with an abstract risk parameter Θ , by variance component models (*additive* random effects models). We find this a very clever and useful reformulation of the credibility theory, but we prefer using *multiplicative* models, since these are common practice in GLM rating.

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To this end, we will restate well-known credibility models as multiplicative random effects models. From a credibility theory point of view, this is just a slight alteration of the Bühlmann-Straub credibility model with *a priori* differences given by Bühlmann & Gisler (2005; Section 4.13, see also Proposition 2.11). Our estimators for this case can also be seen as the non-parametric counterparts of the exact credibility estimators derived in Ohlsson & Johansson (2006a). Since the theory can be found in these sources, we focus on methods rather than proofs in the present text.

Car models are hierarchically ordered under car brands; this suggests using hierarchical credibility models, as noted by Sundt (1987). We present hierarchical credibility estimators in a multiplicative GLMC context, which, to the best of our knowledge, have not appeared in the literature before. These estimators can be brought back on the standard hierarchical credibility case, so once again we focus on practical methods while leaving out the proofs.

The idea of combining credibility and GLMs was introduced by Nelder & Verrall (1997), using the hierarchical GLMs (HGLMs) of Lee & Nelder (1996); their approach uses the concept of hierarchical likelihood, which we presume is not familiar to most actuaries. We prefer here to stay with the distribution-free and simple approach that is standard in credibility theory.

In summary our goal is

- to demonstrate the practical value of combining GLMs with credibility (GLMC), in particular for car model classification;
- give an algorithm for the implementation of these ideas;
- to extend these models to the practically important hierarchical credibility case.

We also mention the pedagogic value we have found in introducing credibility theory for actuarial students via simple multiplicative models, rather than the standard approach with its abstract risk parameters and Bayesian models (see lecture notes in Ohlsson & Johansson (2006b)).

In Section 2, we present the model assumptions, estimators and iterative algorithm for GLMC. Section 3 gives an extension to the hierarchical case. To conclude, Section 4 presents the application to *car model classification* in private motor insurance, with data from Länsförsäkringar Alliance, Sweden.

2. GLMs and credibility theory in multiplicative models

In non-life insurance rating, one works with some *key ratio* $Y = X/w$ of an outcome X to an exposure measure w . For example, Y may be the *risk premium*, with X the observed claim cost and w the number of policy years. In GLMs, it is customary to carry out separate analyses with Y as the *claim frequency* and average *claim severity*, respectively. Our exposition below covers all these cases, but for simplicity, we will mostly mention the risk premium only.

The rating problem consists of estimating how the mean of the risk premium (claim frequency, claim severity) depends on a number of covariates, the rating factors. In particular, in the private lines, we often have a large number of relevant rating factors. For example, in motor insurance these might include the age and gender of the insured person, the age of the car, the annual mileage, etc. Some of these rating factors are categorical with just a few classes, e.g. gender, while others are continuous or ordinal and can be grouped into new variables, such as age class or mileage class.

There are also categorical variables with a large number of levels that cannot be grouped *a priori* by risk homogeneity. One example is *car model*, which has about 2500 levels in Sweden. Even though cars may be grouped by technical variables, such as weight or engine power, typically, there are many residual differences left within such groups (see Section 4). Since data are usually too sparse for the vast majority of models, there is a need for credibility estimators. Furthermore, running a GLM with the 2500 car models cross-classified by all the other rating factors may be impractical. A rating factor such as this will be called a multi-level factor (MLF). Other examples of MLFs are geographical region (defined by e.g. zip code), and the customer herself (experience rating, bonus/malus systems).

We index the key ratio Y_{ijt} , where i refers to a cell in the tariff given by the *ordinary rating factors*, i.e. the non-MLFs. The index j denotes the MLF level, while the number of levels is called J (so that $j=2500$ in the car model case). Repeated observations are indexed by t ; these are either different insurances with the same value for i and j , or annual observations of the same insurance, or both.

The tariff cell i is the cross-tabulation of R different rating factors. We let γ_r^i denote the price relativity for factor number r for insurances in cell i , while μ is the so-called *base premium*, the rating in the *base cell*, in which $\gamma_r^i \equiv 1$, ($r = 1, \dots, R$).

Let u_j be the price relativity for MLF level j , ($j = 1, 2, \dots, J$) measuring the departure of the insurances at that level from their expected values in the tariff of ordinary rating factors. The u_j s cannot be estimated by GLMs, since there are too many levels R . This is a case for using credibility estimators, which can be found by assuming the u_j s are (unobserved observations of) random effects U_j , with $E(U_j) = 1$. The multiplicative model is then

$$E(Y_{ijt}|U_j) = \mu \gamma_1^i \gamma_2^i \dots \gamma_R^i U_j.$$

By initially disregarding U_j , we can estimate μ and the γ_r^i s by standard GLM methods (see, e.g. Brockman & Wright (1992)). Given such estimates, we now look for a credibility estimator of U_j . For simplicity in notation we introduce

$$\gamma_i = \gamma_1^i \gamma_2^i \dots \gamma_R^i.$$

Even though price relativities like U_j are standard in GLM rating, for our derivations it will be more convenient to work with the variable $V_j \doteq \mu U_j$. The multiplicative model can now be written as

$$E(Y_{ijt}|V_j) = \gamma_i V_j. \quad (2.1)$$

In rating with GLMs, it is usually assumed that the data follow a Tweedie model. The most important part of this assumption is that the variance is proportional to the p -th power of

the expectation. In our case, we must condition on the random effect; the Tweedie variance is then

$$\text{Var}(Y_{ijt}|V_j) = \frac{\phi (\gamma_i V_j)^p}{w_{ijt}}.$$

where ϕ is the GLM dispersion parameter. For $p=1$ we get the (overdispersed) Poisson distribution used for the claim frequency; with $p=2$ we get the gamma distribution case often used for claim severity; and for $1 < p < 2$ we have a compound Poisson distribution, suitable for the risk premium (cf. Jørgensen & Paes de Souza (1994)). For more information on Tweedie models, see Jørgensen (1997). In general, it is a very natural assumption in premium rating that the standard deviation of Y is proportional to some power of the mean, and inversely proportional to the exposure weight w .

Assuming that the V_j are identically distributed, we may introduce $\sigma^2 \doteq \phi E[V_j^p]$, independent of j , so that

$$E[\text{Var}(Y_{ijt}|V_j)] = \frac{\gamma_i^p \sigma^2}{w_{ijt}}. \quad (2.2)$$

This is really the only fact about the variance of Y_{ijt} that we need here. Note in particular that we do *not* assume that the data follow a Tweedie distribution, cf. the use of quasi-likelihood in GLMs.

We now restate the usual assumptions in credibility models (see Bühlmann & Gisler (2005) Chapter 4.13 for a relevant version) in our GLM context.

ASSUMPTION 2.1. (a) *The MLF groups are independent, i.e. (Y_{ijt}, V_j) and $(Y_{i'j't'}, V_{j'})$ are independent as soon as $j \neq j'$.*

(b) *The V_j ($j=1, 2, \dots, J$) are identically distributed with $E[V_j] = \mu > 0$ and we write $\tau^2 \doteq \text{Var}[V_j]$.*

(c) *For any j , conditional on V_j , the Y_{ijt} s are mutually independent, with mean given by Eq. (2.1) and variance satisfying Eq. (2.2).*

Next, we transform the observations so that we can bring this situation back to the classical Bühlmann-Straub model. Let

$$\tilde{Y}_{ijt} = \frac{Y_{ijt}}{\gamma_i}; \quad \tilde{w}_{ijt} = w_{ijt} \gamma_i^{2-p}. \quad (2.3)$$

PROPOSITION 2.1. *Under Assumption 2.1, the credibility estimator of V_j is*

$$\hat{V}_j = z_j \bar{\tilde{Y}}_{\cdot j} + (1 - z_j) \mu, \quad (2.4)$$

where

$$z_j = \frac{\tilde{w}_{j\cdot}}{\tilde{w}_{j\cdot} + \sigma^2 / \tau^2}, \quad (2.5)$$

and

$$\bar{\tilde{Y}}_{\cdot j} = \frac{\sum_{i,t} \tilde{w}_{ijt} \tilde{Y}_{ijt}}{\tilde{w}_{\cdot j}} = \frac{\sum_{i,t} \tilde{w}_{ijt} Y_{ijt} / \gamma_i}{\sum_{i,t} \tilde{w}_{ijt}}. \quad (2.6)$$

with $\tilde{Y}_{ijt}$ and $\tilde{w}_{ijt}$ defined in Eq. (2.3).

Consequently,

$$\hat{U}_j = z_j \frac{\bar{\tilde{Y}}_{\cdot j}}{\mu} + (1 - z_j), \quad (2.7)$$

and the rating for insurances with MLF level j in *a priori* tariff cell i is $\mu \gamma_i \hat{U}_j$.

Ohlsson & Johansson (2006a) derived the estimator in Proposition 2.1 as a so called *exact credibility* estimator, namely by assuming that the observations are drawn from a Tweedie GLM and the random effect follows the natural conjugate prior distribution. The proposition above can be seen as a non-parametric counterpart of that result. The proof is omitted since it is close to the proof in Bühlmann & Gisler (2005, Chapter 4.13) for the Bühlmann-Straub estimator with *a priori* information; note, though, that in their setting, all insurances in group j have the same *a priori* information, and that they only consider the case $p = 1$. The reader interested in a detailed proof in our case is referred to Ohlsson (2006).

Note. When Y_{ijt} is *claim frequency*, with w_{ijt} as the number of insurance years, we use the Poisson distribution ($p = 1$) in a GLM, so that $\tilde{w}_{ijt} = w_{ijt} \gamma_i$, a quantity that is called ‘normalized insurance years’ in Campbell (1986). Then

$$\frac{\bar{\tilde{Y}}_{\cdot j}}{\mu} = \frac{\sum_{i,t} w_{ijt} Y_{ijt}}{\sum_{i,t} w_{ijt} \mu \gamma_i},$$

i.e. the number of claims in group j divided by the expected number of claims in the same group – a very natural estimator of U_j . In the case when all insurances in group j have the same γ_i , this is the ‘standardized frequency’ of Bühlmann & Gisler (2005), where the estimator of $\hat{U}_j$ for this case is derived in Corollary 4.9.

For the case when Y_{ijt} is *claim severity* and w_{ijt} is the number of claims, the standard approach is to use a GLM with $p = 2$, corresponding to a gamma distribution. Here

$$\frac{\bar{\tilde{Y}}_{\cdot j}}{\mu} = \frac{\sum_{i,t} w_{ijt} Y_{ijt} / (\mu \gamma_i)}{\sum_{i,t} w_{ijt}},$$

i.e. a weighted average of the observed relative deviance of Y_{ijt} from its expectation $\mu \gamma_i$, again a simple and natural estimator of U_j . $\square$

Thus far, we have assumed that μ and γ_i are known *a priori*. Recall that γ_i is the product of the price relativities $\gamma_1^i, \dots, \gamma_R^i$ in cell i ; these can be estimated by GLMs, treating $\hat{U}_j$ as a known *offset* variable. Therefore, we must know $\hat{U}_j$ to estimate μ and γ_i and vice versa. The solution to this dilemma is to use an iterative procedure as described in Section 2.1.

Note. Traditional texts on credibility are not stated in terms of random effects, but use an abstract random risk parameter Θ_j . The task is then to estimate $\mu(\Theta_j) = E[Y_{jt}|\Theta_j]$. The correspondence to our notation is $\mu(\Theta_j) = V_j$. Since no inference is made on the parameter Θ_j itself, there is no loss in generality in our approach. On the other hand, the random effect case is contained in the traditional model as the case $\mu(\Theta_j) = \Theta$; see Bühlmann & Gisler (2005) for a discussion on some random effect models, in particular for the claim frequency (Poisson) case in Chapter 4.10. However, Bühlmann & Gisler (2005) do not treat the connection to GLMs and Tweedie models. $\square$

2.1. The iterative GLMC algorithm

The above considerations suggest the following algorithm for simultaneous rating of ordinary factors $\gamma_1, \gamma_2, \dots, \gamma_R$ by GLMs and a multi-level factor U_j by credibility, in the multiplicative model $\mu \gamma_1^i \gamma_2^i \dots \gamma_R^i U_j$.

- (0) Initially, let $\hat{U}_j = 1$ for all j .
- (1) Estimate the parameters for the ordinary rating factors by a Tweedie GLM (e.g. Poisson or gamma) with log-link, using $\log(\hat{U}_j)$ as an *offset*-variable. This yields $\hat{\mu}$ and $\hat{\gamma}_1^i, \dots, \hat{\gamma}_R^i$.
- (2) Compute $\hat{\sigma}^2$ and $\hat{\tau}^2$ as given in Section 2.2, using $\hat{\mu}$ and $\hat{\gamma}_1^i, \dots, \hat{\gamma}_R^i$ from Step 1.
- (3) Use Eq. (2.7) to compute $\hat{U}_j$, using the estimates from Steps 1 and 2.
- (4) Return to Step 1 with the new $\hat{U}_j$ from Step 3.

Repeat Steps 1–4 until convergence. In our experience, this is a matter of something similar to three to five iterations, if there are no extreme outliers in the data.

Note. Suppose that for some j we had full credibility, i.e. $z_j = 1$. Then Eq. (2.7) could be rewritten

$$\mu \hat{U}_j = \bar{Y}_{\cdot j} \Leftrightarrow \sum_{i,t} w_{ijt} (\mu \gamma_i)^{2-p} \hat{U}_j = \sum_{i,t} w_{ijt} (\mu \gamma_i)^{1-p} Y_{ijt},$$

by Eqs. (2.3) and (2.6). This could be recognized as the maximum likelihood estimating equation that would result if U_j was just another ordinary rating factor in our GLM.

Now, $z_j = 1$ is just the theoretical limiting case when the exposure tends to infinity. In practice, we might have a large exposure w_{ijt} , by which z_j will be close to 1; then from the above derivation we expect the credibility estimate $\hat{U}_j$ to be close to the GLM maximum likelihood estimate. In this vague sense, the GLM estimate is the limiting case of the credibility estimate when exposure grows. $\square$

2.2. Estimation of variance parameters

An unbiased estimator of τ^2 can be obtained from the corresponding estimator in Bühlmann & Gisler (2005, Section 4.8), while an unbiased estimator for σ^2 can be derived from their Eq. (4.34). Let

$$\hat{\sigma}_j^2 = \frac{1}{n_j - 1} \sum_{i,t} \tilde{w}_{ijt} (\tilde{Y}_{ijt} - \bar{\tilde{Y}}_j)^2. \quad (2.8)$$

Then

$$\hat{\sigma}^2 = \frac{\sum_j (n_j - 1) \hat{\sigma}_j^2}{\sum_j (n_j - 1)}, \quad (2.9)$$

and

$$\hat{\tau}^2 = \frac{\sum_j \tilde{w}_{j\cdot} (\bar{\tilde{Y}}_{j\cdot} - \bar{\tilde{Y}}_{\dots})^2 - (J - 1) \hat{\sigma}^2}{\tilde{w}_{\dots} - \sum_i \tilde{w}_{i\cdot}^2 / \tilde{w}_{\dots}}. \quad (2.10)$$

where $\bar{\tilde{Y}}_{\dots}$ is the $\tilde{w}_{j\cdot}$ -weighted average of the $\bar{\tilde{Y}}_{j\cdot}$ s. Note that in our case these estimators are strictly unbiased only if γ_i is known, while in practice we plug in a GLM estimate here.

3. Hierarchical credibility in multiplicative models

In the example with 2500 *car models*, data indicate that cars of the same brand (Renault, Volvo, etc.) have some risk characteristics in common, even if they represent different models (Renault Megane, Renault Laguna, etc.) (Figure 2). We like to include *brand* as a rating factor in our models; this is of particular importance when a new model of a well-known brand is introduced on the market, with no insurance data available. For some brands (e.g. Volvo) we have plenty data, while for other brands (e.g. Subaru) we have very few insurances in our data files; hence, brand, like car model, is an MLF, for which we want to use credibility estimation.

Since the car models are hierarchically ordered under the brands, this is a case for using *hierarchical credibility*, as suggested by Sundt (1987). As in the previous section, we like to take into account *a priori* information from the standard rating factors (age, gender, mileage, etc.) estimated by GLM and generalize our estimators from Section 2 to the hierarchical case.

Note. Another hierarchical MLF in motor insurance is the geographical zone, where we may have three or more levels, such as zip codes within counties, within states. However, for simplicity, we only consider two-level hierarchical models here. $\square$

Let Y_{ijkt} be the observed key ratio, with denominator (weight) w_{ijkt} , in *a priori* tariff cell i , for *sector* j (car brand) and *group* k (car model) within sector j . Note that while k is hierarchical under j , a particular i can be combined with any j, k and is hence *not* part of the hierarchical ordering. The full multiplicative model is now, with U_j as the random effect for sector and U_{jk} as the random effect for group within sector:

$$E(Y_{ijkt} | U_j, U_{jk}) = \mu \gamma_1^i \gamma_2^i \dots \gamma_R^i U_j U_{jk}.$$

Here we assume $E[U_j] = 1$ and $E[U_{jk} | U_j] = 1$. As before, we use an abbreviated notation for the standard rating factors, $\gamma_i = \gamma_1^i \gamma_2^i \dots \gamma_R^i$. We need credibility estimators of both U_j and

U_{jk} , but again find it easier to work with $V_j = \mu U_j$, and also with $V_{jk} = \mu U_j U_{jk} = V_j U_{jk}$, so that

$$E(Y_{ijkl}|V_j, V_{jk}) = \gamma_i V_{jk}. \quad (3.1)$$

If, given the random effects, Y follows a Tweedie GLM model, then

$$\text{Var}(Y_{ijkl}|V_j, V_{jk}) = \frac{\phi(\gamma_i V_{jk})^p}{w_{ijkl}},$$

where ϕ is the dispersion parameter. This motivates the assumption

$$E[\text{Var}(Y_{ijkl}|V_j, V_{jk})] = \frac{\gamma_i^p \sigma^2}{w_{ijkl}}, \quad (3.2)$$

where $\sigma^2 = \phi E[V_{jk}^p]$. We now generalize Assumption 2.1 to the hierarchical case.

ASSUMPTION 3.1. (a) The sectors are independent, i.e. (Y_{ijkl}, V_j, V_{jk}) and $(Y_{i'j'k' l'}, V_{k'}, V_{j'k'})$ are independent as soon as $j \neq j'$.

(b) For every j , conditional on the sector effect V_j , the groups are independent, i.e. (Y_{ijkl}, V_{jk}) and $(Y_{i'j'k' l'}, V_{j'k'})$ are conditionally independent as soon as $k \neq k'$.

(c) All the pairs (V_j, V_{jk}) , $j = 1, 2, \dots, J$; $k = 1, 2, \dots, K_j$, are identically distributed, with

$$E[V_j] = \mu > 0 \quad \text{and} \quad E(V_{jk}|V_j) = V_j.$$

We use the notation

$$\tau^2 \doteq \text{Var}[V_j] \quad \text{and} \quad v^2 \doteq E[\text{Var}(V_{jk}|V_j)].$$

(d) For any (j, k) , conditional on (V_j, V_{jk}) , the Y_{ijkl} is independent, with mean given by Eq. (3.1) and with variance satisfying Eq. (3.2).

Similar to Eqs. (2.3) and (2.6), we introduce

$$\tilde{Y}_{ijkl} = \frac{Y_{ijkl}}{\gamma_i}, \quad \tilde{w}_{ijkl} = w_{ijkl} \gamma_i^{2-p}, \quad \text{and} \quad \bar{\tilde{Y}}_{\cdot jk \cdot} = \frac{\sum_{i,t} \tilde{w}_{ijkl} \tilde{Y}_{ijkl}}{\sum_{i,t} \tilde{w}_{ijkl}}. \quad (3.3)$$

It is not hard to see that $\tilde{Y}_{ijkl}$ and $\tilde{w}_{ijkl}$ fulfill the assumptions of ordinary hierarchical credibility theory and we obtain the following two propositions (cf. Bühlmann & Gisler (2005, Chapter 6). Direct proofs are given in Ohlsson (2006), partly by using Theorem 1 (ii) of Sundt (1980).

PROPOSITION 3.1. Under Assumption 3.1, the credibility estimator of V_j is

$$\hat{V}_j = q_j \bar{\tilde{Y}}_{\cdot j \cdot}^z + (1 - q_j) \mu, \quad (3.4)$$

where

$$z_{jk} = \frac{\tilde{w}_{\cdot jk \cdot}}{\tilde{w}_{\cdot jk \cdot} + \sigma^2/v^2}, \quad \bar{\tilde{Y}}_{\cdot j \cdot}^z = \frac{\sum_k z_{jk} \bar{\tilde{Y}}_{\cdot jk \cdot}}{\sum_k z_{jk}} \quad \text{and} \quad q_j = \frac{z_{j \cdot}}{z_{j \cdot} + v^2/\tau^2}. \quad (3.5)$$

Note that the mean $\bar{\tilde{Y}}_{\cdot j \cdot}^z$ is weighted by z instead of w . If the group variance v^2 is small then $z_{jk} \approx v^2 \tilde{w}_{\cdot jk \cdot} / \sigma^2$, thereby $q_j \approx \tilde{w}_{\cdot j \cdot} / \tilde{w}_{\cdot j \cdot} + \sigma^2/\tau^2$, so that q_j , as it should, is the credibility factor

in a model with no groups within the sectors, cf. Eq. (2.5). If, on the other hand, $\tau^2 \approx 0$, then $q_j \approx 0$, therefore $\hat{V}_j \approx \mu$, indicating that the sector level may be omitted, as it should when not contributing to the model variance.

PROPOSITION 3.2. *The credibility estimator of V_{jk} is*

$$\hat{V}_{jk} = z_{jk} \bar{Y}_{jk\cdot} + (1 - z_{jk}) \hat{V}_j, \quad (3.6)$$

where $\bar{Y}_{jk\cdot}$, z_{jk} and $\hat{V}_j$ are given in Eqs. (3.3), (3.5) and (3.4), respectively.

By the simple relation between the U 's and V 's, we finally note that the random effects in the multiplicative model $\mu \gamma_1^i \gamma_2^i \dots \gamma_R^i U_j U_{jk}$ can be estimated as follows:

$$\hat{U}_j = q_j \frac{\bar{Y}_{j\cdot}^z}{\mu} + (1 - q_j), \quad \hat{U}_{jk} = z_{jk} \frac{\bar{Y}_{jk\cdot}}{\hat{V}_j} + (1 - z_{jk}).$$

3.1. Estimation of variance parameters

It remains to estimate the parameters σ^2 , τ^2 and ν^2 . Traditionally, iterative so-called pseudo-estimators were suggested in the literature for the hierarchical case, however, recently, direct, unbiased-type estimators were independently derived by Gisler & Bühlmann (2005, Section 6.6) and Ohlsson (2005). These can easily be extended to the case with *a priori* differences with the following result, where T_{jk} is the number of observations i and t for the group (j,k) , K_j is the number of groups in sector j , and J is the number of sectors.

$$\sigma^2 = \frac{1}{\sum_j \sum_k (T_{jk} - 1)} \sum_j \sum_k \sum_i \sum_t \tilde{w}_{ijkt} (\tilde{Y}_{ijkt} - \bar{Y}_{jk\cdot})^2; \quad (3.7)$$

$$\nu^2 = \frac{\sum_j \sum_k \tilde{w}_{jk\cdot} (\bar{Y}_{jk\cdot} - \bar{Y}_{j\cdot})^2 - \hat{\sigma}^2 \sum_j (K_j - 1)}{\tilde{w}_{\dots} - \sum_j \left(\sum_k \tilde{w}_{jk\cdot}^2 \right) / \tilde{w}_{j\cdot}}; \quad (3.8)$$

$$\hat{\tau}^2 = \frac{\sum_j z_j (\bar{Y}_{j\cdot}^z - \bar{Y}_{\dots}^z)^2 - \hat{\nu}^2 (J - 1)}{z_{\dots} - \sum_j z_j^2 / z_{\cdot\cdot}}. \quad (3.9)$$

where $\bar{Y}_{jk\cdot}$ is given by Eq. (3.3), $\bar{Y}_{j\cdot}^z$ by Eq. (3.5), and

$$\bar{Y}_{j\cdot} = \frac{\sum_k \tilde{w}_{jk} \bar{Y}_{jk\cdot}}{\sum_k \tilde{w}_{jk}} \quad \text{and} \quad \bar{Y}_{\dots}^z = \frac{\sum_j z_j \bar{Y}_{j\cdot}^z}{\sum_j z_j}.$$

4. Application to car model classification

In this section, we present some of the application to *car model classification* in car hull insurance, using data from Länsförsäkringar Alliance, a Swedish insurance company. The

car model (such as Renault Scénic 1.8, Volvo V70 2.4) is an important rating factor in motor insurance. As mentioned in Section 2, car model is a MLF with approximately 2500 car models in a preliminary classification.

We also have a large number of ordinary rating factors, such as sex, age, annual mileage etc., and a specific car model j may appear in any *a priori* tariff cell i constructed from the ordinary factors. The presence of both ordinary rating factors and a MLF makes the GLMC algorithm of Section 2.1 appropriate for car model classification.

As is standard in GLM rating, we make separate analyses of *claim frequency* using a Poisson distribution (or rather a variance function μ^p with $p=1$) and *average claim severity* using a gamma distribution (or rather $p=2$). Here we present only the results from the claim frequency part of the study.

The idea to use credibility theory in car model classification was discussed by Campbell (1986) and Sundt (1987). The credibility models of these authors resemble ours; however, these texts were written before GLM became a standard rating tool, therefore their analysis is on *risk premium* – not claim frequency and severity – with weights corresponding to the case $p=1$ in Eq. (2.3), and they do not iterate as we suggest in Section 2.1.

An application of the algorithm in Section 1 with all the ordinary rating factors and car model j gave the result shown by the non-filled bars (with the legend ‘No auxiliaries’) in the histogram in Figure 1, where the bins are formed by rounding the $\hat{U}_j$ s to the nearest first decimal number. We find that almost 30% of the car models received $\hat{U}_j$ s in the span

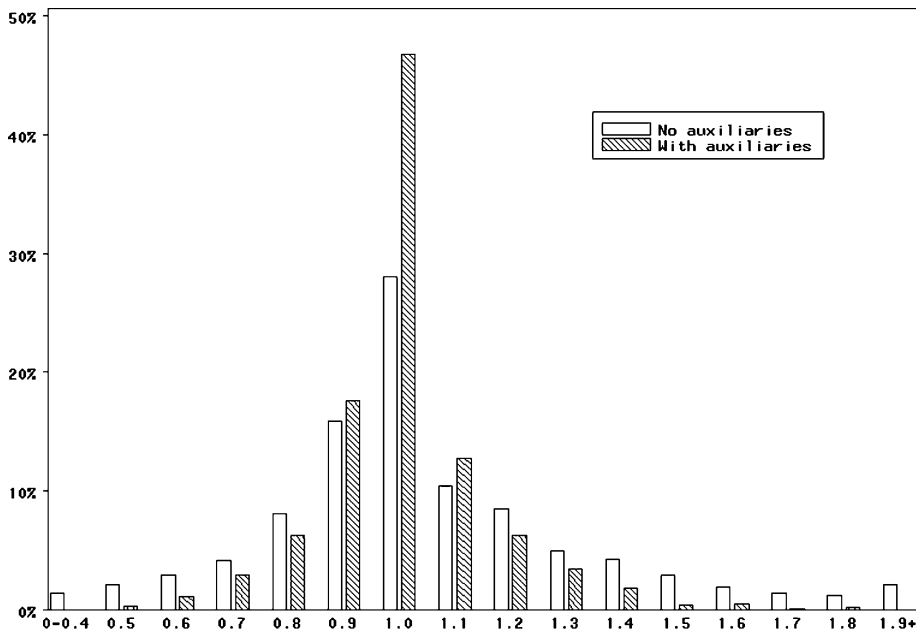


Figure 1. Histogram of credibility estimators $\hat{U}_j$ for car model, with and without auxiliary classification factors in the model.

0.95–1.05, but in most cases, the car model really makes a difference: the extremes differ by a factor of about $1.9/0.4 = 4.75$ from each other.

Another approach to car model classification is to introduce a number of ordinary rating factors that describe the car model technically, such as the weight, the power of the engine, the type of car (saloon, estate car, convertible), etc. We call these variables *auxiliary classification factors*, since they are not intended for separate use in the tariff, but only to help in the car model classification. Let us call this *the factor method* of car model classification; it has the advantage of not requiring large amounts of data for individual car models – a new model can be rated with the same accuracy as an old model.

It is actually possible to have the best of both worlds, by combining the credibility method and the factor method as follows. Add the auxiliary classification factors to the ordinary factors in the GLM analysis and then again use the algorithm in Section 2.1. The resulting $\hat{U}_j$ s are shown in Figure 1 (with the legend ‘With auxiliaries’). Even though over 45% of the car models now get $\hat{U}_j$ s close to 1, there is still a considerable variation among them; this shows that the factor method leaves a lot of residual variation, which can be estimated as a (residual) random effect U_j . A possible explanation is that differences in claim frequency is, to a large extent, a question of drivers behavior and that different models attract different kinds of drivers, not only due to technical differences.

On the other hand, the introduction of the auxiliary factors reduces the variation of the $\hat{U}_j$ s. This is an indication that we get a better tariff from the inclusion of auxiliary factors in our credibility car model classification: a brand new or rare car model will get $\hat{U}_j = 1$ because of lack of data, but the auxiliaries still provide some information on this car, thus reducing the non-explained variation in claim frequency. However, for models with large amounts of data, the introduction of auxiliary variables makes no difference.

This effect is shown in some detail in Table 1, for a sample of car models sorted by the exposure weights $w_{.j}$ (number of policy years). As expected, with large $w_{.j}$, the weighted averages of the observations $\bar{Y}_{.j}$ produce reliable estimates that are given high weight in the credibility formula, and the rating of car models is hardly affected by the introduction of auxiliaries, as seen by comparing the ‘No auxiliaries’ $\hat{U}_j$ to the ‘With auxiliaries’ column c_j , where the product of the auxiliaries’ price relativities is multiplied by the new $\hat{U}_j$ from this round (note that the auxiliaries are tied to the car model j and not to the tariff cell i).

At the other end of the table, with data from very few policy years, credibility is low and the $\bar{Y}_{.j}$ s are shaky. Here, to some extent, one must rely on the auxiliary factors. Without them, the classification would have been much closer to 1 for these cars.

Note. In applications we might have more than one MLF, in motor insurance, besides car model, we may use a geographical area defined by zip code or parish, for example. In principle, the algorithm can be extended to the case with two or more independent MLFs in a rather straightforward fashion; in practice we find it easier to rate one MLF at a time. $\square$

Table 1. Selected car models j with number of policy years $w_{.j}$, weighted averages $\bar{Y}_{.j}$, credibility estimators $\hat{U}_j$ and credibility factors z_j ; without and with auxiliary classification factors; the classification variable c_j is $\hat{U}_j$ multiplied by the price relativities for the auxiliaries.

j	$w_{.j}$	No auxiliaries			With auxiliaries			
		$\bar{Y}_{.j}$	$\hat{U}_j$	z_j	$\bar{Y}_{.j}$	$\hat{U}_j$	z_j	c_j
1	41 275	0.74	0.74	1.00	0.98	0.98	0.99	0.75
2	39 626	0.58	0.58	1.00	0.89	0.89	0.99	0.59
3	39 188	0.59	0.59	1.00	0.86	0.86	0.99	0.60
4	31 240	0.82	0.82	1.00	0.93	0.93	0.99	0.82
5	28 159	0.49	0.50	1.00	0.74	0.75	0.98	0.50
401	803	2.08	1.95	0.88	1.43	1.35	0.82	1.99
402	802	0.97	0.97	0.86	1.11	1.08	0.70	0.95
403	801	1.77	1.66	0.86	1.54	1.40	0.74	1.62
404	799	0.74	0.78	0.86	0.83	0.88	0.69	0.79
405	798	1.32	1.27	0.86	0.73	0.78	0.82	1.41
901	181	1.38	1.22	0.58	1.14	1.06	0.42	1.29
902	180	1.61	1.38	0.63	0.91	0.95	0.56	1.70
903	180	2.28	1.76	0.59	1.35	1.18	0.51	2.01
904	179	0.79	0.88	0.56	0.86	0.95	0.34	0.88
905	179	2.38	1.80	0.58	1.52	1.25	0.48	1.98
1801	7	2.39	1.07	0.05	2.05	1.03	0.03	1.22
1802	7	4.63	1.19	0.50	3.86	1.08	0.03	1.31
1803	7	0.00	0.96	0.04	0.00	0.99	0.01	0.55
1804	7	0.00	0.95	0.05	0.00	0.98	0.02	0.87
1805	7	0.00	0.94	0.06	0.00	0.98	0.02	0.58

4.1. The hierarchical case

As discussed at the beginning of Section 3, the *car brand* might contain valuable information not contained in the auxiliary (technical) data on the car. Now, car brand is itself an MLF, and, following Sundt (1987), we try a hierarchical credibility model for this situation. In our case, the multiplicative model includes: first the ordinary rating factors, then the auxiliary factors based on technical properties of the car models, then the car brand stochastic effect, and finally the car model stochastic effect (cf. Table 2, where A denotes the number of auxiliary rating factors).

Note that even though car brand is a random effect in the hierarchical credibility model, from a practical point of view it serves the same purpose as the auxiliaries: to improve the estimation of car model factors by extracting information that is common to several car models, and hence may be estimated more accurately than the car models themselves. The full multiplicative model is then:

$$E(Y_{ijkl}|U_j, U_{jk}) = \mu \gamma_1^i \dots \gamma_R^i \gamma_{R+1}^{jk} \dots, \gamma_{R+A}^{jk} U_j U_{jk}.$$

Table 2. Variables in car model classification with GLMC, hierarchical case.

Rating factors	Example	Notation
Ordinary rating factors	Age of driver	$\gamma_1^i, \gamma_2^i, \dots, \gamma_R^i$
Auxiliary factors	Engine power	$\gamma_{R+1}^{jk}, \gamma_{R+2}^{jk}, \dots, \gamma_{R+A}^{jk}$
Car brand	Volvo	U_j
Car model	Volvo V70 2.4	U_{jk}

This model can be estimated using a straightforward hierarchical credibility extension of the algorithm in Section 2.1, using the estimators from Section 3. Figure 2 displays $\hat{U}_j$ for the 96 car brands; we can see that there are substantial differences between many of the 96 car brands, but not as large as between car models. We conclude that it is worthwhile including the car brand level in our multiplicative model; this is especially important for the classification of car models on which we have none or few data.

After analyzing claim frequency and claim severity by the above method, we multiply the auxiliary factors, the car brand factor $\hat{U}_j$, and the car model factor $\hat{U}_{jk}$, from each of these two analyses and get

$$c_{jk} = \gamma_{R+1}^{jk} \cdots \gamma_{R+A}^{jk} \hat{U}_j \hat{U}_{jk}.$$

The c_{jk} s for claim frequency and severity are then multiplied, giving a classification variable for the risk premium. The classification is finalized by calibrating this variable to maintain the total premium level, as demonstrated in Sundt (1987, Section 3.4), and then forming car model classes from the calibrated variable.

Our overall conclusion is that the GLMC algorithm in Section 2.1 is an efficient tool for car model classification and that the results are improved by the augmentation of the model with auxiliary technical variables as well as the car brand, estimated by the hierarchical credibility estimators from Section 3.

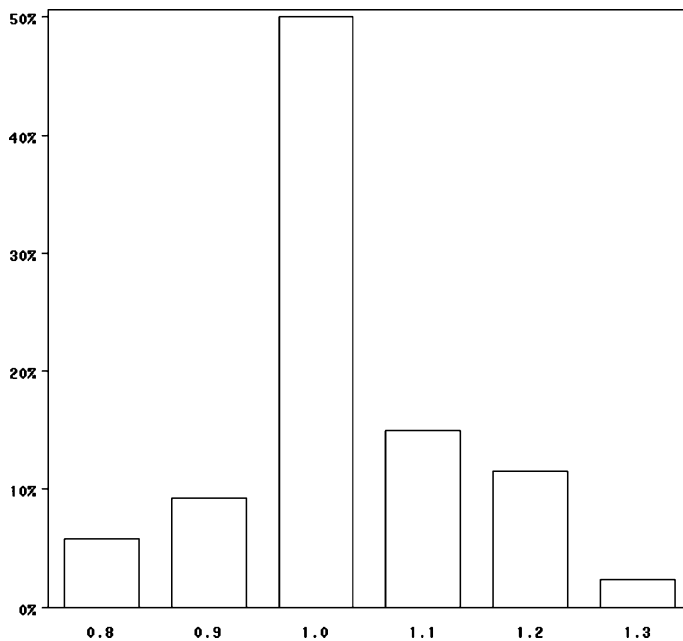


Figure 2. Histogram of credibility estimators $\hat{U}_j$ for car brands.

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